

5. Requirement Appendix

This section provides additional requirement information for the FoodResQ system, including common business rules, shared system requirements, application messages, and other supporting requirements.

5.1 Business Rules

Common business rules that the FoodResQ system must follow are listed below.

ID	Rule Definition
BR-01	Only registered and active users can access role-based features in the system.
BR-02	A user can register as Provider, Receiver, Charity Organization, or Volunteer depending on their selected role.
BR-03	A Provider must have an active account before creating food listings.
BR-04	Food listings must include required information such as title, category, quantity, pickup location, pickup time, expiry time, and image.
BR-05	Food listing expiry time must be in the future at the time of creation.
BR-06	Food listing status follows the lifecycle: Draft > Available/Active > Partially Reserved > Fully Reserved > Completed/Expired/Cancelled.
BR-07	A listing cannot be edited or deleted if it already has confirmed reservations, except for allowed non-critical fields where permitted.
BR-08	The system automatically marks listings as expired when the expiry time has passed.
BR-09	Receivers can search food listings by location, category, availability, and expiry time.
BR-10	Receivers can reserve only available listings with sufficient remaining quantity.
BR-11	Reservation quantity must not exceed the remaining quantity or the maximum quantity allowed per reservation.
BR-12	Reservation creation must be processed atomically to prevent multiple users from reserving more food than available.
BR-13	A receiver can cancel a reservation only before pickup or before the configured cancellation deadline.
BR-14	Provider and Receiver can confirm pickup through QR code or manual confirmation depending on system support.
BR-15	Completed reservations can be used for rating, feedback, and trust score updates.
BR-16	Users who repeatedly cancel late, no-show, or violate food safety rules may lose trust score points.
BR-17	Users with low trust score may be restricted or banned according to system trust score rules.
BR-18	Volunteer Shippers must set availability and location before receiving delivery offers.
BR-19	Delivery offers are sent only to eligible nearby Volunteer Shippers.
BR-20	The first Volunteer Shipper who accepts a delivery offer is assigned to the delivery task.

ID	Rule Definition
BR-21	A Volunteer Shipper can handle only one active delivery task at a time.
BR-22	Delivery status must follow the correct order: pending assignment > assigned > heading to provider > quality checked > in transit > delivered.
BR-23	Failed or cancelled delivery tasks require a reason before the system updates the status.
BR-24	Charity Organizations can create kitchen campaigns only when their organization profile is valid.
BR-25	Campaigns must include basic information such as title, description, schedule, kitchen address, expected servings, and required volunteer roles.
BR-26	Campaign status follows the lifecycle: Draft > Open > In Progress > Completed / Cancelled.
BR-27	Volunteers can apply only for campaign roles that match their specialization, such as Chef, Server/Waiter, or Shipper.
BR-28	Campaign volunteer assignments must be reviewed and approved before volunteers can access operation tasks.
BR-29	Volunteer Chefs can update cooking tasks and food safety logs only when they are assigned to the campaign.
BR-30	Volunteer Servers/Waiters can scan handoff QR codes only when they are assigned to the distribution event.
BR-31	Each receiver handoff QR code can be used only once per valid distribution round.
BR-32	Campaign experience can be submitted only by volunteers who joined the campaign.
BR-33	Users can submit reports for food issues, delivery issues, user behavior, or campaign problems.
BR-34	Admin can review reports and apply warning, suspension, or resolution actions.
BR-35	Admin can manage users, listings, reports, campaigns, volunteer verification, and system statistics.
BR-36	Uploaded images must be valid JPG, PNG, or WEBP files and must not exceed the configured size limit.
BR-37	Sensitive user data, authentication tokens, and personal information must be protected by secure transmission and authorization checks.
BR-38	Users can view only their own private records unless they have admin-level permission.
BR-39	Notifications are sent for important events such as reservation confirmation, delivery assignment, campaign approval, report status, and pickup reminders.
BR-40	All system timestamps must be stored and processed consistently using server time and returned in ISO date format.

5.2 Common Requirements

Common requirements that apply across the system are listed below.

ID	Requirement Definition
CR-01	The system must provide role-based access control for Admin, Provider, Receiver, Charity Organization, Volunteer Shipper, Volunteer Chef, and Volunteer Server/Waiter.
CR-02	The system must require authentication before accessing protected screens and APIs.
CR-03	The system must validate all required input fields before submitting data to the backend.
CR-04	The system must display clear error messages when validation fails.
CR-05	The system must prevent unauthorized users from accessing data that does not belong to them.
CR-06	The system must support responsive web UI for desktop and mobile browser sizes.
CR-07	The system must support image upload for food listings, user avatars, campaign images, proof images, and verification evidence.
CR-08	The system must provide fallback images when uploaded images are missing or unavailable.
CR-09	The system must provide loading states while waiting for API responses.
CR-10	The system must provide empty states when no data is available.
CR-11	The system must show toast messages for successful or failed actions.
CR-12	The system must support pagination or limited result loading for lists such as listings, reservations, reports, and history.
CR-13	The system must protect APIs with JWT access tokens and refresh tokens.
CR-14	The system must automatically clear expired sessions and redirect users to login when authentication fails.
CR-15	The system must use secure password hashing before saving user passwords.
CR-16	The system must support real-time or near-real-time notifications for important events.
CR-17	The system must support location-based listing search and delivery matching.
CR-18	The system must prevent race conditions when multiple receivers reserve the same listing.
CR-19	The system must log or store important operational records such as reservations, deliveries, reports, campaign assignments, and trust score changes.
CR-20	The system must allow Admin to monitor key platform metrics and operational activities.

ID	Requirement Definition
CR-21	The system must allow users to update their profile information.
CR-22	The system must allow users to report abnormal or unsafe situations.
CR-23	The system must ensure uploaded files are validated by type and size before saving.
CR-24	The system must support QR-based confirmation for pickup, handoff, or delivery-related workflows where applicable.
CR-25	The system must keep completed, cancelled, failed, and expired records for history and reporting purposes.

5.3 Application Messages List

The table below lists common application messages displayed in validation, toast notifications, empty states, and system errors.

#	Message Code	Message Type	Context	Content
1	MSG01	Inline	No search result	No food listings found.
2	MSG02	Inline under field	Required field is empty	This field is required.
3	MSG03	Toast	Login failed	Incorrect email or password. Please try again.
4	MSG04	Toast	Unauthorized access	You need to login to continue.
5	MSG05	Toast	Register successfully	Account registered successfully. Please verify your account.
6	MSG06	Toast	Duplicate email or phone	Email or phone number already exists.
7	MSG07	Inline under field	Invalid phone number	Phone number is not valid.
8	MSG08	Inline under field	Weak password	Password must contain at least 8 characters, uppercase letter, lowercase letter, number, and special character.
9	MSG09	Toast	Create listing successfully	Food listing created successfully.
10	MSG10	Toast	Create listing failed	Failed to create food listing. Please try again.
11	MSG11	Inline under field	Expiry time invalid	Expiry time must be in the future.
12	MSG12	Toast	Invalid image upload	Invalid image. Only JPG, PNG, or WEBP files under 5MB are allowed.

#	Message Code	Message Type	Context	Content
13	MSG13	Toast	Update listing successfully	Food listing updated successfully.
14	MSG14	Toast	Delete listing successfully	Food listing deleted successfully.
15	MSG15	Toast	Cannot edit reserved listing	This listing cannot be edited because it already has reservations.
16	MSG16	Toast	Cannot delete reserved listing	This listing cannot be deleted because it already has reservations.
17	MSG17	Toast	Reserve food successfully	Food reserved successfully.
18	MSG18	Toast	Reserve food failed	Failed to reserve food. Please try again.
19	MSG19	Toast	Insufficient quantity	Not enough quantity remaining for this listing.
20	MSG20	Toast	Reservation cancelled	Reservation cancelled successfully.
21	MSG21	Toast	Pickup confirmed	Pickup confirmed successfully.
22	MSG22	Toast	Invalid QR code	Invalid or expired QR code.
23	MSG23	Toast	Rating submitted	Rating submitted successfully.
24	MSG24	Toast	Report submitted	Report submitted successfully.
25	MSG25	Inline	No reservation history	No reservation history found.
26	MSG26	Toast	Delivery offer accepted	Delivery task accepted successfully.
27	MSG27	Toast	Delivery offer rejected	Delivery offer rejected.
28	MSG28	Toast	Offer expired	This delivery offer has expired.
29	MSG29	Toast	Delivery status updated	Delivery status updated successfully.
30	MSG30	Toast	Invalid delivery status	Delivery status cannot be updated in this order.
31	MSG31	Toast	Delivery failed	Delivery marked as failed.
32	MSG32	Inline under field	Missing failure reason	Failure reason is required.
33	MSG33	Toast	Availability updated	Availability status updated successfully.
34	MSG34	Toast	Location permission denied	Location permission is required to receive nearby delivery tasks.
35	MSG35	Toast	Campaign created	Campaign created successfully.
36	MSG36	Toast	Campaign updated	Campaign updated successfully.

#	Message Code	Message Type	Context	Content
37	MSG37	Toast	Campaign cancelled	Campaign cancelled successfully.
38	MSG38	Toast	Apply campaign successfully	Campaign application submitted successfully.
39	MSG39	Toast	Duplicate campaign application	You have already applied for this campaign role or shift.
40	MSG40	Toast	Campaign full	This campaign role or shift is already full.
41	MSG41	Toast	Volunteer approved	Volunteer assignment approved successfully.
42	MSG42	Toast	Volunteer rejected	Volunteer assignment rejected successfully.
43	MSG43	Toast	Cooking task updated	Cooking task updated successfully.
44	MSG44	Toast	Safety log submitted	Food safety log submitted successfully.
45	MSG45	Toast	Handoff recorded	Meal handoff recorded successfully.
46	MSG46	Toast	Duplicate handoff	This receiver has already been recorded for this distribution round.
47	MSG47	Toast	Feedback submitted	Feedback submitted successfully.
48	MSG48	Toast	Experience submitted	Campaign experience submitted successfully.
49	MSG49	Toast	Admin action completed	Admin action completed successfully.
50	MSG50	Toast	Network error	Network error. Please check your connection and try again.
51	MSG51	Toast	Server error	Server error. Please try again later.
52	MSG52	Inline	No notification	No notifications yet.
53	MSG53	Toast	Session expired	Your session has expired. Please login again.
54	MSG54	Toast	File upload failed	File upload failed. Please try again.
55	MSG55	Inline	No active task	No active task available.

5.4 Other Requirements

Other supporting requirements and quality expectations are listed below.

ID	Other Requirement
OR-01	The system should be easy to use for non-technical users such as food receivers, charity staff, and volunteers.
OR-02	The web interface should maintain consistent navigation, color, typography, and layout across all user roles.
OR-03	The system should provide meaningful fallback UI when data is loading, empty, or unavailable.
OR-04	Uploaded images should be displayed through the API media server or cloud storage URL.
OR-05	The system should avoid exposing internal server errors directly to end users.
OR-06	Admin screens should provide clear status badges for users, listings, reports, campaigns, and volunteer assignments.
OR-07	The system should support Vietnamese user-facing text for operational screens.
OR-08	The system should preserve historical records for reporting, auditing, and trust score calculation.
OR-09	The system should support future migration from local upload storage to cloud storage such as S3, Cloudflare R2, or Bunny CDN.
OR-10	The system should support future expansion for mobile app workflows including QR scanning, GPS location, delivery tracking, and campaign participation.
OR-11	The system should provide clear confirmation dialogs before destructive actions such as deleting, cancelling, banning, or rejecting.
OR-12	The system should ensure that dashboard statistics are calculated from completed or valid records only.
OR-13	The system should allow configuration of key rules such as reservation limits, search radius, trust score thresholds, and QR validity time.
OR-14	The system should maintain separation between public pages, authenticated user pages, and admin management pages.
OR-15	The system should be able to handle common failure cases such as expired QR code, missing image file, failed upload, failed network request, and invalid token.